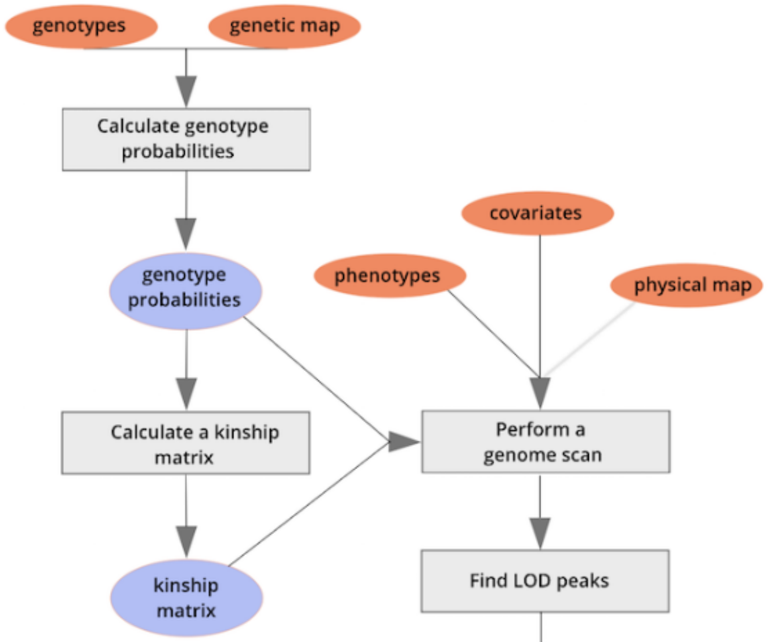
Pheno2GeneRank

Introduction: Biological Background of GWAS or QTL



Candidate Gene List



Trait / Phenotype

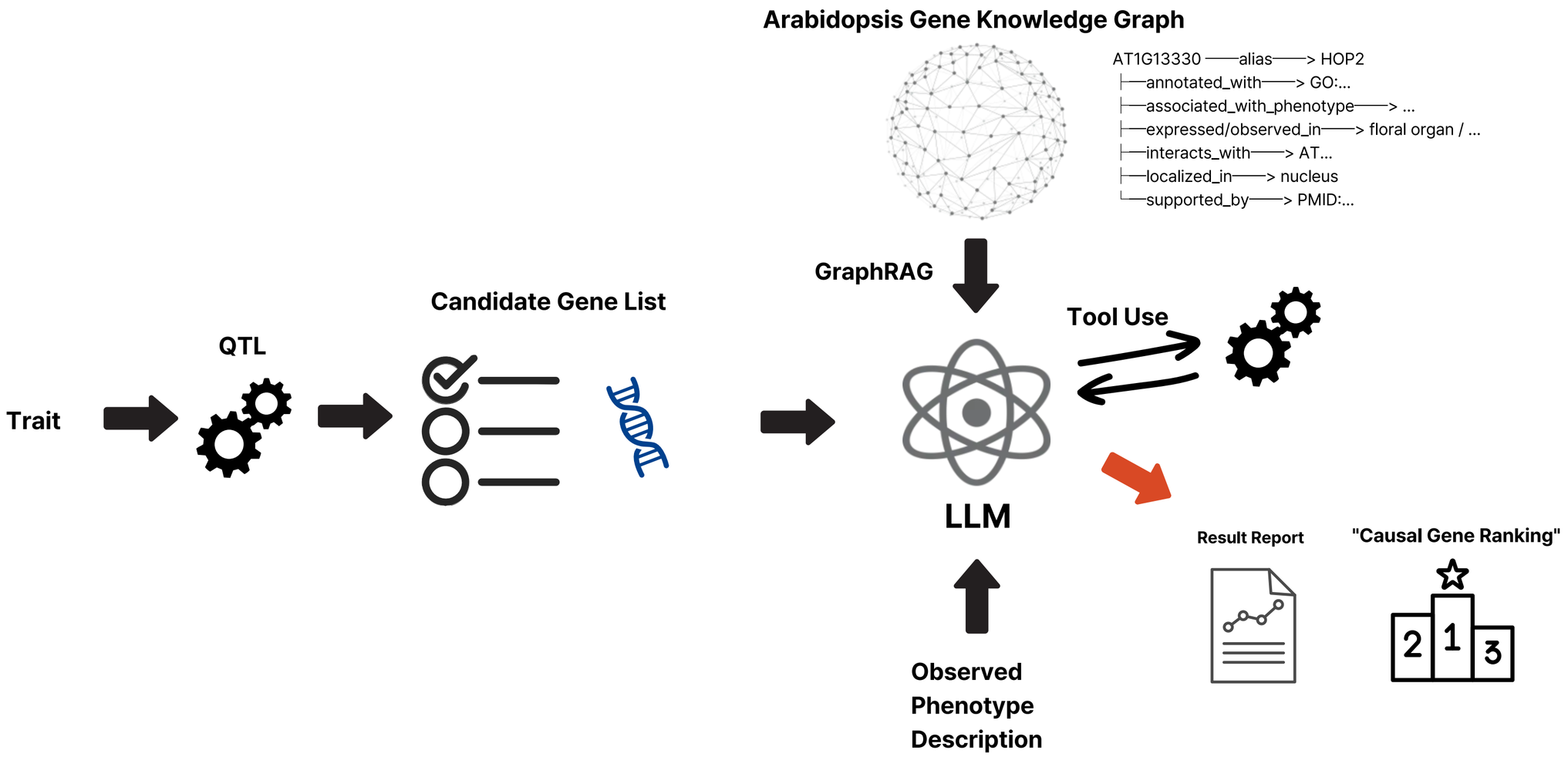






Gene Exploration & Selection

Idea of Pheno2GeneRank: Causal Gene Ranking LLM Agent through Observed Phenotype Descriptions



Evaluation of Pheno2GeneRank: Previous tools (QTG_Finder, QTG-LGBM)

Feature Engineering
→ Machine Learning

Use only predefined features

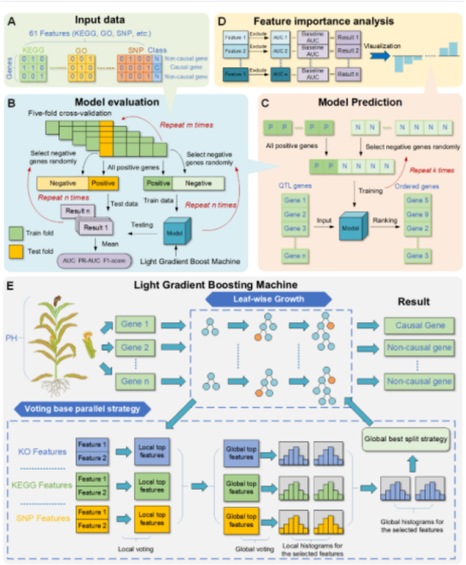
Do not use
trait / phenotype context

QTG_Finder (version 2.0)

QTG-Finder is a machine-learning pipeline to prioritize causal genes for QTLs identified by linkage mapping. We trained QTG-Finder models for Arabidopsis, rice, sorghum, Setaria viridis based on known causal genes and orthologs of known causal genes, respectively. By utilizing additional information like polymorphisms, function annotation, co-function network, paralog copy number, the models can prioritize causal genes for QTLs identified by QTL mapping.

Authors: Fan Lin, February 2020
Environment: Python=3.7.12 pandas=0.25.3 numpy=1.19.5 scikit-learn=0.21.2

QTG-LGBM: a method of prioritizing causal genes in quantitative trait loci in maize



vs

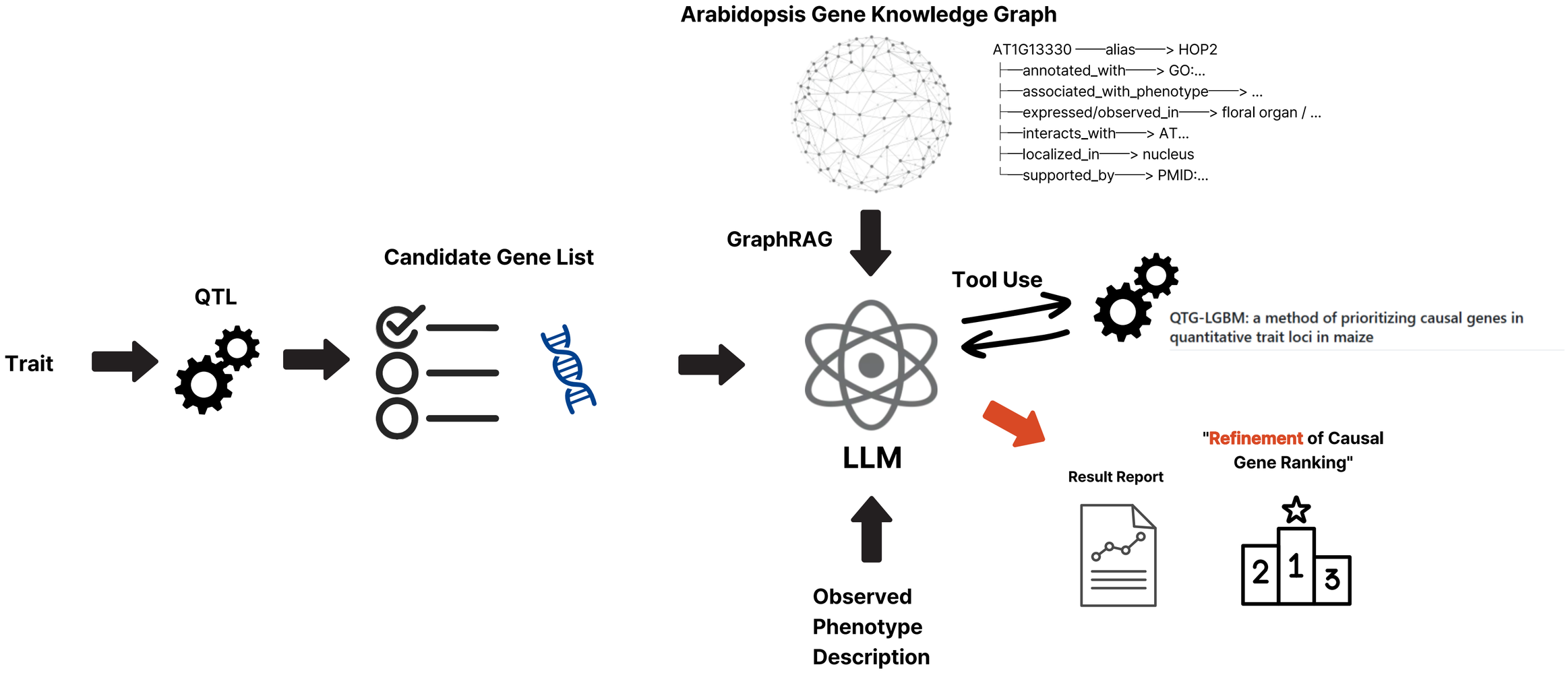
Pheno2GeneRank

Can use trait / phenotype context

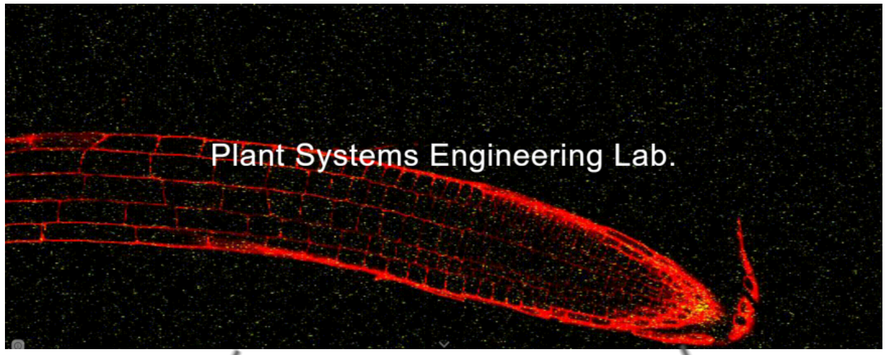
Return User-Friendly Report

Idea 2 of Pheno2GeneRank: Use QTG-LGBM as a tool for Pheno2GeneRank

→ We can use both the predefined features (that are used for QTG-LGBM) and phenotype context



"Cowork"



Supervision
for Pheno2GeneRank

Plant System Research with Pheno2GeneRank



Systems Biology and Bioinformatics Laboratory

Pheno2GeneRank Development

1. Algorithm Design
2. Ablation Study

Evaluation

1. Metric Design
2. Test Dataset Construction
3. Validation